Simultaneous indicators of two entire functions of order two

GPT-6 Astra

Abstract

Let a finite collection of pairwise independent linear forms on $\mathbb{C}^2$ be assigned individually admissible indicators of order two. We prove that the indicators are simultaneously realizable precisely when the maximum of every two assigned indicators is the maximum of the whole collection. The construction allows arbitrary continuous indicators and does not impose completely regular growth. Its main step is a Hermitian metric whose directional weights are the prescribed indicators and whose curvature is positive after a correction of size $o(|z|^2)$. Weighted $\bar\partial$ estimates provide peak sections, and a Baire argument makes all prescribed indicators exact.

1 Statement and background

For a nonzero entire function of order two and finite type, its indicator is

$$h_f(\theta) = \limsup_{r\to\infty} r^{-2}\log|f(re^{i\theta})|.$$

Normal type means positive finite type. We identify a row vector $a = (a_1, a_2)$ with the linear form $v \mapsto av = a_1v_1 + a_2v_2$. A continuous 2π-periodic function h is 2-*trigonometrically convex* if $h'' + 4h$ is a nonnegative distribution. Equivalently, $|z|^2 h(\arg z)$, extended by zero at the origin, is subharmonic. The classical indicator theory gives continuity and this convexity condition for finite-type indicators, and realizes every such admissible indicator; see Bernstein [1].

Theorem 1.1. *Let $A \subset \mathbb{C}^2$ be finite, contain $(1,0)$ and $(0,1)$, and have every two of its elements linearly independent. For each $a \in A$ let h_a be a continuous 2π-periodic 2-trigonometrically convex function with positive maximum. Put $H = \max_{a\in A} h_a$. There exist entire functions f_1, f_2 such that every $a_1f_1 + a_2f_2$ has order two, normal type, and indicator h_a if and only if*

$$\max(h_a(\theta), h_b(\theta)) = H(\theta) \qquad (a \neq b,\ \theta \in \mathbb{R}). \tag{1}$$

Necessity is immediate: any two forms in A are a basis of the dual space, so every other combination is a constant linear combination of them. The indicator of a sum is at most the maximum of the summands' indicators. We prove sufficiency.

Eremenko [3] poses the simultaneous indicator problem for linear combinations and identifies (1) as the natural condition for two functions. He explains its sufficiency at order one through Laplace transforms. Generalized transforms involve ramified indicator surfaces; Maergoiz [4] gives a scalar continuation theorem in that setting. The argument here works directly with growth weights on the plane and resolves the stated finite-family, order-two case. It uses the vector-bundle L^2 existence theorem for $\bar\partial$ in Demailly [2, Theorem 6.1].

No regular-growth limit is required. All growth equalities below concern the directional limsup defining the indicator.

2 Deficit intervals and finite approximations

Write

$$u_a(re^{i\theta}) = r^2 h_a(\theta), \qquad U(re^{i\theta}) = r^2 H(\theta).$$

These functions are subharmonic. Condition (1) says that at each angle at most one h_a is strictly less than H. Thus the open sets

$$D_a = \{\theta : h_a(\theta) < H(\theta)\}$$

are pairwise disjoint. Their connected components, regarded as arcs of the circle, will be called deficit intervals and labelled by a.

Lemma 2.1. *The functions h_a and H are Lipschitz. For each integer $k \geq 1$ there are 2-trigonometrically convex functions $h_a^{[k]}$ with only finitely many deficit intervals in total such that*

$$h_a \leq h_a^{[k+1]} \leq h_a^{[k]} \leq h_a + 2^{-k}, \qquad \max(h_a^{[k]}, h_b^{[k]}) = H \quad (a \neq b). \tag{2}$$

Each deficit interval of the approximating family is an entire original deficit interval, with its original profile.

Proof. On a sufficiently short angular interval, divide $h(\theta)$ by a positive function $\cos 2(\theta - \theta_0)$ and use the coordinate $\tan 2(\theta - \theta_0)$. Trigonometrical convexity becomes ordinary convexity. Local Lipschitz continuity and finite one-sided derivatives follow, and compactness of the circle gives a common Lipschitz bound for our finite family and its maximum.

Retain every deficit interval whose height $\sup_I(H - h_a)$ exceeds 2^{-k}; set $h_a^{[k]} = h_a$ on retained intervals and $h_a^{[k]} = H$ elsewhere. A common Lipschitz bound gives a positive lower bound on the width of every interval of height greater than 2^{-k}. There are therefore only finitely many retained intervals.

We check convexity at their endpoints. At a left endpoint α, contact and $h_a \leq H$ give

$$H'_-(\alpha) \leq h'_{a,-}(\alpha) \leq h'_{a,+}(\alpha).$$

At a right endpoint β they give

$$h'_{a,-}(\beta) \leq h'_{a,+}(\beta) \leq H'_+(\beta).$$

Consequently pasting an original interval into H creates nonnegative derivative jumps. If two retained intervals of the same label meet at an endpoint, the original derivative jump is retained. The differential inequality holds in each piece and at every junction. The remaining assertions follow directly from the construction. A deficit occupying the whole circle is treated as one interval without endpoints. □

Fix a nonnegative smooth radial convolution kernel χ supported in the unit disk, with integral one. Denote the convolutions of $r^2 h_a^{[k]}(\theta)$ and U by $W_a^{[k]}$ and V. They are smooth subharmonic functions, and there is a constant independent of k such that

$$W_a^{[k]}(z) = r^2 h_a^{[k]}(\theta) + O(r+1), \qquad V(z) = r^2 H(\theta) + O(r+1), \tag{3}$$

$$|\nabla W_a^{[k]}(z)| + |\nabla V(z)| \leq C(r+1), \qquad W_a^{[k]} \leq V. \tag{4}$$

Indeed the original homogeneous potentials have gradients bounded almost everywhere by $C|z|$, and convolution preserves this bound with $|z|$ replaced by $|z| + 1$. If two original potentials agree on a sector, their convolutions agree at points of that sector at distance greater than one from its boundary. This exact agreement will specify all metric overlaps.

3 A shear calculation for Hermitian metrics

For a smooth positive Hermitian matrix Q on the plane, use the curvature endomorphism

$$K_Q = -\partial_{\bar{z}}(Q^{-1}\partial_z Q). \tag{5}$$

It is self-adjoint in the metric Q. A scalar factor e^{-2u} contributes $2u_{z\bar{z}}\mathrm{Id}$ to curvature. Inequalities between curvature endomorphisms will always use their metric quadratic forms.

Lemma 3.1. *In fixed coordinates (x, y) consider the metric*

$$q = e^{-2u}|x|^2 + e^{-2v}|y + c(z)x|^2.$$

Put $d = v - u$ and $w = e^{-2d}$. In a metric-orthonormal frame its curvature differs from $\mathrm{diag}(2u_{z\bar{z}}, 2v_{z\bar{z}})$ *by an operator of norm at most*

$$w|c_{\bar{z}}|^2 + \sqrt{w}(|c_{z\bar{z}}| + 2|d_z|\,|c_{\bar{z}}|). \tag{6}$$

Proof. At the point under consideration subtract the constant value of c from y, a constant holomorphic change of frame. At that point $c = 0$ and $Q = \mathrm{diag}(e^{-2u}, e^{-2v})$. Differentiate

$$Q = \begin{pmatrix} e^{-2u} + e^{-2v}|c|^2 & e^{-2v}\bar{c} \\ e^{-2v}c & e^{-2v} \end{pmatrix}$$

in (5). After metric normalization the result is

$$\begin{pmatrix} 2u_{z\bar{z}} - w|c_{\bar{z}}|^2 & -\sqrt{w}(\bar{c}_{z\bar{z}} - 2d_{\bar{z}}\bar{c}_z) \\ -\sqrt{w}(c_{z\bar{z}} - 2d_z c_{\bar{z}}) & 2v_{z\bar{z}} + w|c_{\bar{z}}|^2 \end{pmatrix}. \tag{7}$$

The norm of its off-diagonal part is the modulus of either off-diagonal entry, and the remaining error is diagonal with entries of opposite signs. This proves (6). □

We will keep the more restrictive row a fixed and move a complementary row b. Choose b_* with $\det(a, b_*) = 1$. Multiplying b by a scalar of modulus one does not change $|bv|^2$, so it can be written

$$b = s(b_* + ca), \qquad s > 0. \tag{8}$$

The factor s is absorbed by replacing the high weight v with $v - \log s$. Bounded paths of complements, with determinants bounded away from zero, give bounded s, s^{-1}, c. Angular derivatives of $\log s$ change scalar curvature by $O(r^{-2})$. All constants associated with angular derivatives may depend on a finite template's interval widths.

4 Constructing the metric

Define the target squared norm on $\mathbb{C}^2$ by

$$N(z, \xi)^2 = \sum_{a \in A} |a\xi|^2 e^{-2u_a(z)}. \tag{9}$$

Inside a deficit interval labelled a, this is uniformly equivalent to

$$|a\xi|^2 e^{-2u_a(z)} + |b\xi|^2 e^{-2U(z)} \tag{10}$$

for bounded complementary rows b with determinants bounded away from zero. With no deficit, it is equivalent to $e^{-2U(z)}|\xi|^2$. These assertions follow from finite-dimensional norm equivalence and the fact that all other weights equal U.

Proposition 4.1. *There is a smooth positive Hermitian metric Q on the trivial rank-two bundle over $\mathbb{C}$ and a function $\varepsilon(r) \to 0$ such that*

$$e^{-\varepsilon(r)r^2} N(z,\xi)^2 \le \xi^* Q(z)\xi \le e^{\varepsilon(r)r^2} N(z,\xi)^2 \qquad (r \ge 1) \tag{11}$$

and

$$K_Q(z) \ge (1+|z|^2)^{-1}\mathrm{Id}. \tag{12}$$

We give the construction in three steps, including the transitions needed for infinitely many deficit intervals.

4.1 Finite angular templates

Fix one finite retained family from Lemma 2.1. Around each endpoint choose a constant angular collar, with different noncoincident endpoints having disjoint collars. At a contact of intervals labelled a, b, use the fixed row frame (a, b) and the metric

$$|a\xi|^2 e^{-2W_a} + |b\xi|^2 e^{-2W_b}. \tag{13}$$

If the neighbouring region is a gap with no retained deficit, use (a, b) for any complementary row b and the weights W_a, V. The collar extends across the original endpoint, including the entire physical convolution strip when r is sufficiently large. If the same label occurs on both sides, use that row and a fixed complement.

Within each deficit interval labelled a, keep a fixed and join the complementary rows prescribed at its endpoints. Formula (8) constructs such a path: interpolate $\log s$ and c. Make its angular variation occur in a compact subinterval of the deficit interval. There the deficit is bounded below by a positive constant. In a gap use $e^{-2V}M(\theta)$, with M a smooth positive matrix joining the endpoint matrices. Convex interpolation suffices in this scalar-equal region.

For all sufficiently large r, the convolved weights agree exactly on each overlap where these pieces are joined. Thus the pieces give one smooth metric. On a shear support, after absorption of s, the estimates are

$$d \ge c_0 r^2 - C, \quad |d_z| \le Cr, \quad |c_{\bar z}| \le C/r, \quad |c_{z\bar z}| \le C/r^2.$$

Lemma 3.1 makes its curvature error exponentially small, apart from the $O(r^{-2})$ scalar correction from s. In a scalar-equal gap the factor e^{-2V} contributes nonnegative curvature and $M(\theta)$ contributes $O(r^{-2})$. The endpoint frames are constant and their weights subharmonic. Consequently a finite template satisfies

$$K \ge -Cr^{-2}\mathrm{Id}. \tag{14}$$

If a deficit occupies the whole circle, one fixed frame gives the metric without any endpoint construction.

All pointwise frame matrices and their inverses may be bounded by constants depending only on A. There are finitely many possible endpoint rows; choose bounded complement paths for each possible pair. Convex combinations of the resulting positive gap matrices retain a uniform positive lower bound. Only derivative bounds depend on interval widths.

4.2 Preparing and activating a finite enlargement

Consider the nested retained families at levels k and $k+1$. New intervals lie in old gaps. We first prepare their frames without changing the old scalar weights, and then activate their deficits. A parameter $t \in [0, 1]$ will subsequently depend on $\log r$; dots denote differentiation with respect to $\log r$.

For preparation, compare the old angular template with the frame template for the enlarged interval family equipped with the old smoothed weights $W_a^{[k]}$. Thus new intervals have scalar-equal weights away from endpoint collars; throughout the collars the old convolved weights are retained, including any contribution from an adjacent old interval of the same label. On an old a-interval, their complementary rows can be represented as

$$b_j = s_j(b_* + c_j a), \qquad s_j > 0, \quad j = 0, 1.$$

Use

$$s_t = \exp((1-t)\log s_0 + t \log s_1), \qquad c_t = (1-t)c_0 + tc_1. \tag{15}$$

Both source and target complements are constant in angle near each endpoint. Choose a common smaller collar there and use (15) throughout that collar on both sides of the endpoint. In particular, this fixed-dominant-row formula covers the convolved unequal-weight strip. A contact of two old labels retains its original fixed frame.

Away from these collars an old gap has exactly equal convolved weights for sufficiently large r. Interpolate its source and target positive matrices there. To match (15), use a slightly larger collar contained in the constant endpoint-matrix collars of both templates, and blend the matrix $a^*a + b_t^* b_t$ with the convex interpolation of the source and target matrices. A smooth angular cutoff performs this blending entirely in the scalar-equal region. At $t = 0$ both terms equal the source matrix on the blending collar; at $t = 1$ both equal the target matrix. The preparation therefore matches the given metrics exactly at both parameter endpoints. No arbitrary interpolation of unequal-weight matrices is used.

For a radial parameter $t = t(\log r)$ with $|\dot{t}|, |\ddot{t}| \le 1$, the extra radial shear derivatives satisfy

$$|\nabla_{\mathrm{rad}} c_t| \le C|\dot{t}|/r, \qquad |(c_t)_{z\bar{z}}| \le C(1 + |\dot{t}| + |\ddot{t}| + |\dot{t}|^2)/r^2.$$

Angular derivatives occur only in old scalar-equal gaps or in compact subsets of old deficits. On the latter, $d \ge c_0 r^2 - C$. On endpoint collars there is no angular frame variation; radial frame variation costs at most $C|\dot{t}|$ in (6), using $|d_z| \le Cr$. The positive factor s_t satisfies the corresponding bounds. Hence preparation gives

$$K \ge -C(r^{-2} + |\dot{t}| + |\ddot{t}|)\mathrm{Id}. \tag{16}$$

Now hold the prepared frame fixed and set

$$W_a(t) = (1-t)W_a^{[k]} + tW_a^{[k+1]}. \tag{17}$$

For constant t these scalar weights are subharmonic. Near every endpoint the frame is fixed and (17) defines both diagonal entries. Inside a new interval, away from its endpoint collars, the low weight is $u = V - tD$, where $D = V - W_a^{[k+1]} \ge 0$, and the high weight is V. On the compact support of angular frame variation,

$$D \ge c_0 r^2, \qquad |\nabla D| \le Cr, \qquad D \le Cr^2$$

for sufficiently large r. Differentiating a radial t adds at most $C(|\dot{t}| + |\ddot{t}|)$ to the negative part of scalar curvature. In the shear estimate, $d = tD$ up to the bounded $\log s$ term, and

$$|d_z| \le Cr(t + |\dot{t}|) + C/r.$$

The key uniform bound, including t close to zero, is

$$te^{-c_0 t r^2} \le \frac{1}{ec_0 r^2} \qquad (0 \le t \le 1). \tag{18}$$

Since $|c_{\bar{z}}| \le C/r$, Lemma 3.1 again gives (16). On old shear supports the weights are unchanged; endpoint collars use the scalar estimate directly. Touching new and old intervals of the same label are covered by the same fixed dominant row and the convex interpolation (17). Thus (16) holds uniformly throughout the enlargement, with one finite constant for that enlargement.

4.3 The infinite construction and curvature correction

Perform the finite preparations and activations on successive disjoint annuli in the coordinate $\log r$. At stage k, choose the starting radius sufficiently large for all geometric conditions above and for $C_k/r^2 \le 2^{-k}$. Choose a fixed smooth cutoff, constant near its endpoints, evaluated at

$$\frac{\log r - L_k}{T_k}.$$

Taking T_k large makes $C_k(|\dot{t}| + |\ddot{t}|) \le 2^{-k}$. Insert stationary annuli when necessary before the next stage. Every stage uses finite constants and finite radii. The resulting metric Q_0 is smooth outside a disk; extend it smoothly and positively across that disk.

Its curvature has a radial lower bound

$$K_{Q_0} \ge -\beta(r)\mathrm{Id}, \qquad \beta(r) \longrightarrow 0.$$

A smooth nonnegative majorant may be used for β. During stage k, every blended scalar weight differs from its original homogeneous weight by at most $2^{-k}r^2 + O(r+1)$. The pointwise frame bounds are uniform. Equations (3) and (10) consequently imply the two-sided comparison

$$\xi^* Q_0(z)\xi = N(z,\xi)^2 \exp(o(r^2)), \tag{19}$$

uniformly in angle and nonzero ξ, where this notation denotes two-sided multiplicative bounds.

Choose a smooth nonnegative radial function $\eta(r) \to 0$ with

$$\eta(r) \ge \beta(r) + (1+r^2)^{-1},$$

and put

$$\phi(r) = 4\int_0^r \frac{1}{t}\int_0^t s\eta(s)\,ds\,dt.$$

Choose η smooth as a radial function at zero. Then $\phi_{z\bar{z}} = \eta$ and $\phi(r) = o(r^2)$. The latter follows by splitting the inner integral at a fixed radius and then using an arbitrarily small upper bound for η beyond that radius. Finally set $Q = e^{-2\phi}Q_0$. Its curvature is $K_{Q_0} + 2\eta\mathrm{Id}$, which satisfies (12); its comparison (11) follows from (19). This proves Proposition 4.1.

Remark 4.2. The determinant is consistent with this construction. Put $m = \min_a h_a$. Then $H + m$ is 2-trigonometrically convex. For each finite retained family this is locally a sum of its two neighbouring scalar weights, or $2H$ in a gap. Uniform convergence of the finite approximations proves the statement for the original family, including accumulating deficit intervals.

5 Peak sections

We use the following standard form of the vector-valued L^2 estimate. If a smooth Hermitian metric G on the trivial bundle over $\mathbb{C}$ satisfies $K_G \ge \lambda(z)\mathrm{Id} > 0$, then a compactly supported smooth $(0,1)$-form g has a solution $\bar{\partial}v = g$ with

$$\int_{\mathbb{C}} v^* G v\,dA \le C\int_{\mathbb{C}} \lambda^{-1} g^* G g\,dA. \tag{20}$$

The harmless numerical constant depends on the convention for area and form norms. This is Theorem 6.1 of [2], applied on the complete Euclidean plane in bidegree $(1,1)$; identify the resulting $(1,0)$-form with its coefficient using dz. Positivity in this one-dimensional base is precisely the endomorphism inequality used above.

Let $\mathcal{H}$ be the Hilbert space of entire column vectors $F = (f_1, f_2)^T$ with

$$\|F\|_{\mathcal{H}}^2 = \int_{\mathbb{C}} F^* Q F \, (1 + |z|^2)^{-1} \, dA < \infty. \tag{21}$$

It is a closed subspace of the weighted L^2 space: local positive lower bounds on Q give local L^2 convergence, and the distributional equation $\bar\partial F = 0$ is closed.

Lemma 5.1. *For $a \in A$ and $z_0 = Re^{i\theta}$, let $L_{a,z_0}(F) = aF(z_0)$. As $R \to \infty$,*

$$\log \|L_{a,z_0}\| = u_a(z_0) + o(R^2), \tag{22}$$

uniformly in a and θ.

Proof. Set $\delta = (1+R)^{-1}$. On $B(z_0, \delta)$, the Lipschitz bounds on homogeneous potentials imply $u_a(z) = u_a(z_0) + O(1)$. The submean inequality for the scalar entire function aF, followed by (11), gives

$$|aF(z_0)|^2 \le C\delta^{-2}(1+R^2) e^{2u_a(z_0)+o(R^2)} \|F\|_{\mathcal{H}}^2.$$

This proves continuity of evaluation and the upper estimate.

For the lower estimate choose $v \in \mathbb{C}^2$ with

$$av = e^{u_a(z_0)}, \qquad N(z_0, v) \le C_A. \tag{23}$$

If a covector b is uniquely deficient at θ and $a \neq b$, take a fixed vector in $\ker b$ on which a is one, multiplied by $e^{u_a(z_0)} = e^{U(z_0)}$. If $a = b$, use a bounded right inverse of a multiplied by $e^{u_a(z_0)}$; all other weights are larger. If there is no deficit, use any bounded right inverse of a. The finiteness and pairwise independence of A give a uniform constant in (23).

Let p be v times a smooth cutoff supported in $B(z_0, \delta)$ and equal to one on $B(z_0, \delta/2)$. For $\epsilon > 0$ apply (20) to $g = \bar\partial p$ and the metric

$$G_\epsilon(z) = \frac{Q(z)}{|z - z_0|^2 + \epsilon^2}.$$

The extra scalar factor adds the nonnegative curvature of $\log(|z - z_0|^2 + \epsilon^2)$, so the lower bound $(1 + |z|^2)^{-1}$ is preserved. On the support of g, the singular factor and cutoff derivatives are bounded by powers of $1 + R$, uniformly for small ϵ. Moreover (11), the local Lipschitz bound, and (23) give

$$v^* Q(z) v \le e^{o(R^2)} N(z_0, v)^2 \le C_A^2 e^{o(R^2)}.$$

The estimate therefore gives corrections v_ϵ with a uniformly bounded singular weighted norm, at most a polynomial in $1 + R$ times $e^{o(R^2)}$.

Let ϵ decrease to zero. Weak compactness on compact sets and a diagonal subsequence give a distributional solution v_0 of $\bar\partial v_0 = \bar\partial p$ with

$$\int_{\mathbb{C}} \frac{v_0^* Q v_0}{|z - z_0|^2} \, dA \le e^{o(R^2)}. \tag{24}$$

Here and below polynomial factors are absorbed into $e^{o(R^2)}$. The correction is holomorphic near z_0, and (24) forces $v_0(z_0) = 0$. Also

$$\frac{|z - z_0|^2}{1 + |z|^2} \le 2(1 + R^2),$$

so (24) controls its norm in (21). The cutoff p has the same required norm bound. Hence $F = p - v_0$ is entire,

$$F(z_0) = v, \qquad \|F\|_{\mathcal{H}} \le e^{o(R^2)}.$$

Its evaluation in (23) proves the lower estimate in (22). □

6 Exact realization and an example

Every $F \in \mathcal{H}$ satisfies the uniform upper growth estimate

$$\log |aF(re^{i\theta})| \le r^2 h_a(\theta) + o(r^2) + \log \|F\|_{\mathcal{H}} \tag{25}$$

by the evaluation upper bound. To obtain exact indicators, fix a countable dense set T of angles. For $a \in A$, $\theta \in T$, positive rational ϵ, and positive integer n, define the open set

$$\mathcal{U}(a,\theta,\epsilon,n) = \bigcup_{\substack{R\in\mathbb{N}\\ R\ge n}} \left\{ F : \ |aF(Re^{i\theta})| > e^{(h_a(\theta)-\epsilon)R^2} \right\}.$$

It is dense. Indeed, in any norm ball choose R sufficiently large that the lower evaluation estimate in Lemma 5.1, multiplied by the ball's radius, exceeds the displayed threshold. Perturb the centre along a vector nearly attaining the evaluation norm, choosing its complex phase to reinforce the centre's evaluation. The resulting vector lies in the ball and in $\mathcal{U}$.

Baire's theorem supplies F belonging to all these sets. Its combinations have indicator at least h_a at every angle of T, and (25) gives the reverse inequality everywhere. A dense angle at which $h_a > 0$ shows that each combination has order exactly two and positive type; its type is finite by (25). Continuity of its order-two indicator now extends equality from T to the whole circle. Since the coordinate covectors belong to A, the two coordinates of F have the required order and type. This completes the proof of Theorem 1.1.

For an example that does not reduce to functions of z^2, let

$$I_j = [j\pi/2, (j+1)\pi/2], \qquad j = 0,1,2,3,$$

and define the periodic functions

$$b_j(\theta) = \begin{cases} \sin(2(\theta - j\pi/2)), & \theta \in I_j, \\ 0, & \theta \notin I_j. \end{cases}$$

At both endpoints of I_j the derivative jumps upward by two, so $b_j'' + 4b_j$ is a positive measure. Take four pairwise independent forms, for example $(1,0), (0,1), (1,1), (1,2)$, and prescribe

$$H = \sum_{j=0}^{3} b_j = |\sin 2\theta|, \qquad h_j = H - b_j.$$

Each h_j is a sum of three trigonometrically convex functions with positive maximum. Their deficits have disjoint interiors, so the maximum condition holds. The theorem realizes all four indicators, although each h_j fails to be π-periodic. The substitution $f(z) = g(z^2)$ would force that periodicity.

7 Verification and scope

The argument is analytic. Its principal construction consists of the finite-template curvature estimate, the radial transitions, and the subquadratic correction in Proposition 4.1. Lemma 5.1 uses the established vector-valued L^2 theorem with an explicit logarithmic singularity; it does not assume pre-existing nonzero sections in the Hilbert space.

The accompanying exact-arithmetic verifier checks the four entries of the local shear identity (7) by direct matrix differentiation. This verifies that algebraic identity only. The global metric construction, the L^2 argument, and Baire realization are proved in the text. No formal proof-assistant verification or external peer review is claimed.

The theorem concerns a finite family of forms on $\mathbb{C}^2$ at order two. It asserts ordinary indicators and normal type, without asserting completely regular growth.

References

[1] V. Bernstein, *Sulle proprietà caratteristiche delle indicatrici di crescenza delle trascendenti intere d'ordine finito*, Memorie della Reale Accademia d'Italia, Classe di Scienze Fisiche, Matematiche e Naturali, 7. Serie (1936), 131–189.

[2] J.-P. Demailly, *Applications of the theory of L^2 estimates and positive currents in algebraic geometry*, Grenoble summer-school notes, June–July 2007, Theorem 6.1, p. 39. `https://www-fourier.univ-grenoble-alpes.fr/~demailly/manuscripts/eem2007.pdf`.

[3] A. Eremenko, *Indicators of linear combinations*, 21 July 2026, Problem 1. `https://www.math.purdue.edu/~eremenko/dvi/indi2.pdf`.

[4] L. S. Maergoiz, *The multivalent indicator and conjugate diagrams of an entire function of order $\rho \neq 1$. Application to the solution of algebraic equations*, Journal of Mathematical Sciences **261** (2022), 792–807; Russian original, Zap. Nauchn. Sem. POMI **491** (2020), 94–118. `https://doi.org/10.1007/s10958-022-05789-w`.